

Place and Route of the 16-bit Fibonacci Generator

Taking the Lab 2 netlist through IC Compiler

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The assignment

Run the IC Compiler flow from the tutorial on a design from the previous lab: floorplan, power plan, placement, clock tree synthesis and routing. Generate and analyze the reports after CTS and after routing.

At a glance

	Lab 2 (DC, area run)	After CTS	After routing
Library corner	typ	max	max
Wires	none	estimated	routed + extracted
Clock	5 ns	5 ns	5 ns
Critical path	4.35 ns	[ns]	[ns]
Setup slack	+0.56 ns	[ns]	[ns]
Violating paths	0	[]	[]
Hold violations		[]	[]
Cells	122	[]	[]
Utilization		[%]	[%]
Open nets / DRCs			[] / []

Table 1: Lab 2 synthesis next to the two ICC checkpoints.

1 The design

FIBO from Lab 2: three 16-bit registers (48 flip-flops) and one shared adder that produce 1, 1, 2, 3, 5, 8, ... Ports: clk, rst, en and OUT[15:0], 19 signal pins.

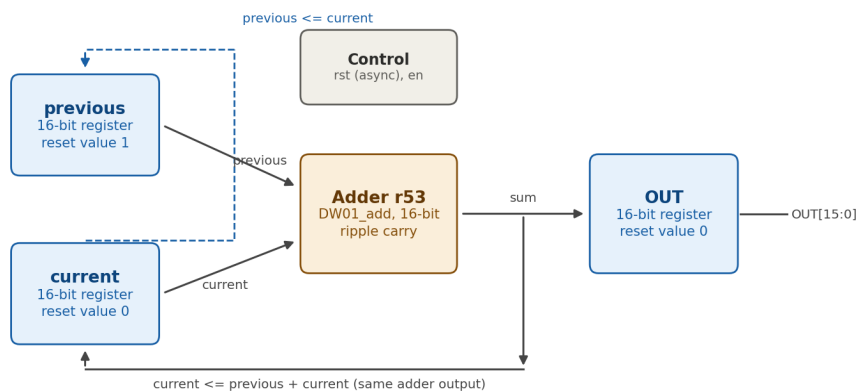


Figure 1: Block diagram from Lab 2.

2 Script

Same flow as the tutorial with new names (FIBO_LIB, top FIBO, fibo_area.sdc, checkpoints FIBO_init, FIBO_cts_opt, FIBO_route). One real difference: the Lab 2 script for this design never wrote a .ddc, so the Verilog netlist is imported instead.

```
import_designs "../output/fibo_area_synth.v" -format verilog -top "FIBO" -cel "FIBO"
read_sdc ../constraints/fibo_area.sdc
```

A Verilog netlist has no constraints inside it, so read_sdc is what brings in the 5 ns clock.

3 Routed layout

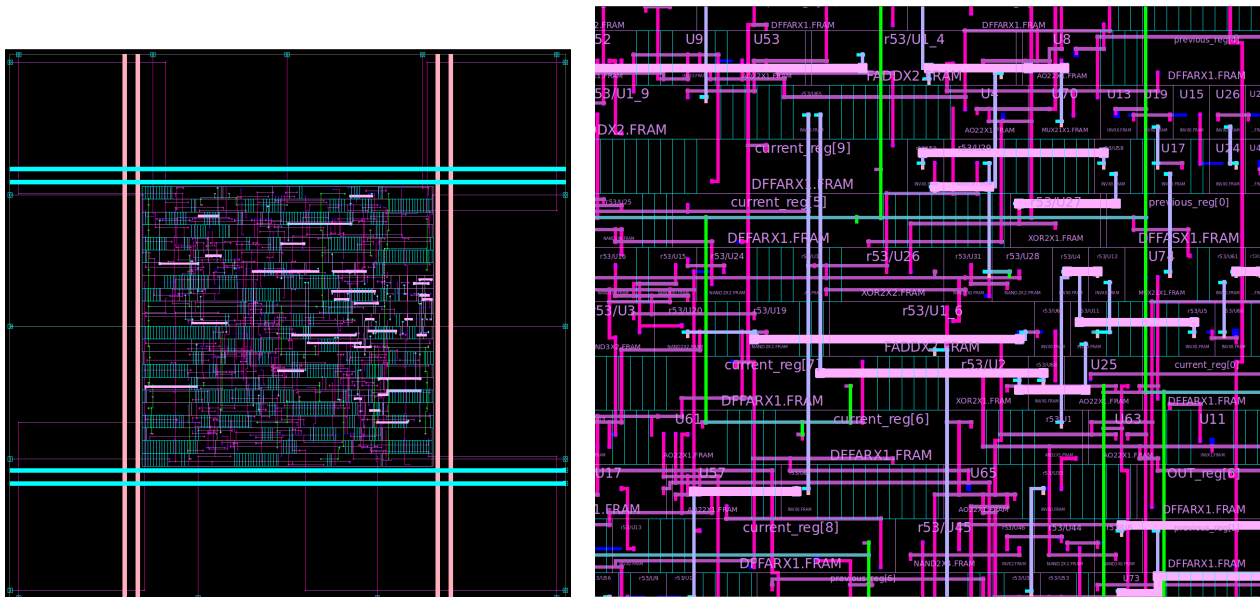


Figure 2: Left: the routed block. Right: zoom on the core: adder stages r53/U1_4, r53/U1_6, r53/U1_9, the current_reg and previous_reg flip-flops and OUT_reg[6].

- **Clock net.** Clicking a clock wire gives net clk, type Clock, layer M4, width $0.16\mu\text{m}$: after clock_opt the clock is a real routed net, not an ideal signal any more.
- **Reset value shows in the cells.** previous resets to 1, so previous_reg[0] is a set-type DFFASX1 while the other flip-flops are DFFARX1, the same mix Lab 2 reported.
- **The adder keeps its name.** The full adders are still r53/U1_1 to r53/U1_15, so the Lab 2 critical path can be followed cell by cell in the layout.

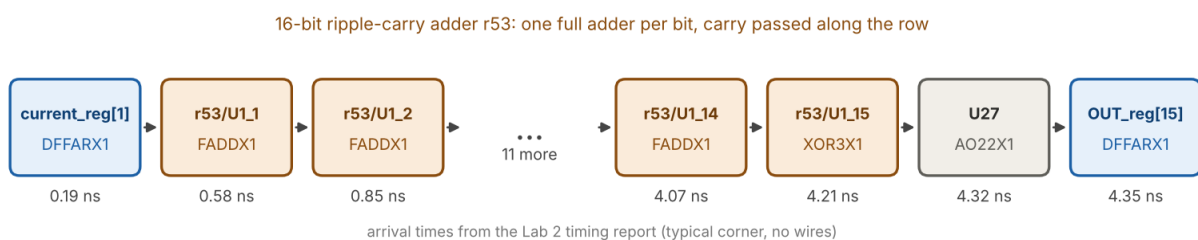


Figure 3: The Lab 2 critical path. Every box is a cell that can be found by name in the routed layout.

4 Reports

Report	What I checked	Result
FIB0_route_qor.rpt	slack, violating paths	[]
FIB0_route_setup.rpt	worst setup path	[start → end, slack]
FIB0_route_hold.rpt	worst hold path	[]
FIB0_route_util.rpt	standard cell utilization	[]
verify_zrt_route	open nets, DRCs	[]

Table 2: Post-route reports.

Compared with Lab 2

The 0.56 ns of slack in Lab 2 was at the typical corner with no wires. At the slow corner with routed wires the same carry chain gives [ns] of slack, so the design [still meets / misses] 5 ns.

5 Inside the cells

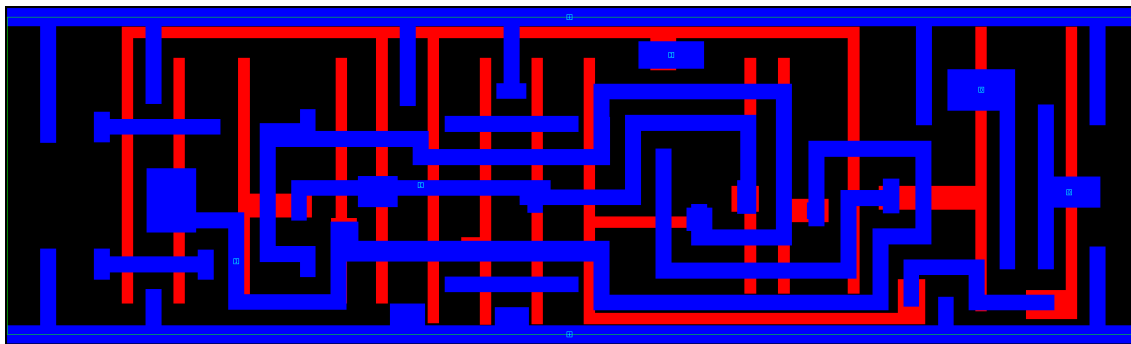


Figure 4: FADDX1, FRAM view: the full adder repeated 14 times in the carry chain. It has five pin squares (A, B, carry in, sum, carry out) and is one of the widest logic cells in the design.

6 What I learned

- A Verilog netlist imports fine, but the constraints must come from the SDC.
- Instance names from synthesis survive place and route, so a timing path from Design Compiler can be traced in the layout.
- Reset values are visible in the hardware: one set-type flip-flop among 47 clear-type ones.

A Source files

RTL: src/assignment2.sv

```
'timescale 1ns/10ps

module FIB0 ( input  logic    clk,
              input  logic    rst,
              input  logic    en,
```

```

        output logic [15:0] OUT );

    logic [15:0] previous;
    logic [15:0] current;

    always_ff @(posedge rst, posedge clk)
    begin
        if (rst) begin
            OUT      <= 16'b0;
            current  <= 16'b0;
            previous <= 16'b1;
        end
        else if (en) begin
            previous <= current;
            current  <= current + previous;
            OUT      <= current + previous;
        end
    end
end
endmodule

```

Testbench: src/assignment2_tb.sv

```

// Testbench for fibonacci generator (module assignment2.sv)
`timescale 1ns/10ps

module assignment2_tb;

    logic      clk;
    logic      rst;
    logic      en;
    logic [15:0] OUT;

    FIB0 dut ( .clk(clk), .rst(rst), .en(en), .OUT(OUT) );

    initial clk = 0;
    always #1 clk = ~clk;

    initial begin
        $monitor("Time=%0t | rst=%b en=%b | OUT=%0d ", $time, rst, en, OUT);
        rst = 1;
        en  = 0;
        @(posedge clk);
        rst = 0;
        en  = 0;
        @(posedge clk);
        en  = 1;
        repeat(5) @(posedge clk); // f(0) through f(4)
        en  = 0;
        @(posedge clk);          // hold steady
        en  = 1;
        repeat(2) @(posedge clk); // f(5), f(6)
        rst = 1;
        en  = 0;
        @(posedge clk);
        $display("Test finished.");
        $finish;
    end
endmodule

```

ICC script: src/icc_fibo.tcl

```

# icc_fibo.tcl : Lab 3, Assignment 1 (16-bit Fibonacci generator, top = FIB0)
# Lab 2 wrote no .ddc for this design, so the Verilog netlist is imported instead.
# Run from icc_assignment1/, one level below output/ and constraints/:
#   icc_shell -shared_license -f icc_fibo.tcl | tee icc_fibo.log

gui_start

set_app_var search_path /apps/designlib/SAED90_EDK/SAED_EDK90nm/Digital_Standard_cell_Library/
synopsys/models/
set_app_var target_library "saed90nm_max.db"
set_app_var link_library "* saed90nm_max.db"

create_mw_lib -technology "/apps/designlib/SAED90_EDK/SAED_EDK90nm/Technology_Kit/milkyway/
saed90nm_icc_1p9m.tf" -mw_reference_library "/apps/designlib/SAED90_EDK/SAED_EDK90nm/
Digital_Standard_cell_Library/process/astro/fram/saed90nm/" "FIB0_LIB"

open_mw_lib FIB0_LIB

import_designs "../output/fibo_area_synth.v" -format verilog -top "FIB0" -cel "FIB0"

read_sdc ../constraints/fibo_area.sdc

set_tlu_plus_files -max_tluplus "/apps/designlib/SAED90_EDK/SAED_EDK90nm/Technology_Kit/starrcxt/
tluplus/saed90nm_1p9m_1t_Cmax.tluplus" -min_tluplus "/apps/designlib/SAED90_EDK/SAED_EDK90nm/
Technology_Kit/starrcxt/tluplus/saed90nm_1p9m_1t_Cmin.tluplus" -tech2itf_map "/apps/designlib/
SAED90_EDK/SAED_EDK90nm/Digital_Standard_cell_Library/process/astro/tech/tech2itf.map"

save_mw_cel -as FIB0_init

create_floorplan -core_utilization 0.6 -start_first_row -left_io2core "30" -bottom_io2core "30" -
right_io2core "30" -top_io2core "30"

derive_pg_connection -power_net "VDD" -power_pin "VDD" -ground_net "VSS" -ground_pin "VSS" -
create_ports "top"

create_rectangular_ring -nets {VSS} -left_offset 0.5 -left_segment_layer M6 -left_segment_width
1.0 -extend_ll -extend_lh -right_offset 0.5 -right_segment_layer M6 -right_segment_width 1.0 -
extend_rl -extend_rh -bottom_offset 0.5 -bottom_segment_layer M7 -bottom_segment_width 1.0 -
extend_bl -extend_bh -top_offset 0.5 -top_segment_layer M7 -top_segment_width 1.0 -extend_tl -
extend_th

create_rectangular_ring -nets {VDD} -left_offset 1.8 -left_segment_layer M6 -left_segment_width
1.0 -extend_ll -extend_lh -right_offset 1.8 -right_segment_layer M6 -right_segment_width 1.0 -
extend_rl -extend_rh -bottom_offset 1.8 -bottom_segment_layer M7 -bottom_segment_width 1.0 -
extend_bl -extend_bh -top_offset 1.8 -top_segment_layer M7 -top_segment_width 1.0 -extend_tl -
extend_th

create_power_strap -nets {VSS} -layer M6 -direction vertical -width 3
create_power_strap -nets {VDD} -layer M6 -direction vertical -width 3

create_fp_placement

place_opt
clock_opt

save_mw_cel -as FIB0_cts_opt

report_placement_utilization > ../output/FIB0_cts_util.rpt
report_qor > ../output/FIB0_cts_qor.rpt
report_timing -delay max -max_paths 5 > ../output/FIB0_cts.setup.rpt
report_timing -delay min -max_paths 5 > ../output/FIB0_cts.hold.rpt

```

```
insert_stdcell_filler -cell_with_metal "SHFILL2" -cell_without_metal "SHFILL2" -connect_to_power {
    VDD} -connect_to_ground {VSS}

route_opt
verify_zrt_route

report_placement_utilization > ../output/FIB0_route_util.rpt
report_qor > ../output/FIB0_route_qor.rpt
report_timing -delay max -max_paths 5 > ../output/FIB0_route.setup.rpt
report_timing -delay min -max_paths 5 > ../output/FIB0_route.hold.rpt

extract_rc -coupling_cap
write_parasitics -format SBPF -output ../output/FIB0.output.sbpf
write_verilog ../output/FIB0_LIB.output.v

save_mw_cel -as FIB0_route
close_mw_cel
```